

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df=pd.read_csv(r"C:\Users\STUDENTS\Downloads\Housing price dataset - Sheet1.csv")
print(df)
```

	Order_ID	Customer_ID	Gender	Category	Product	Quantity	Price	\
0	1001	C001	Male	Electronics	Laptop	1	55000	
1	1002	C002	Female	Clothing	T-Shirt	3	800	
2	1003	C003	Male	Grocery	Rice Bag	2	1500	
3	1004	C004	Female	Furniture	Chair	4	2500	
4	1005	C005	Male	Electronics	Mobile	2	18000	
5	1006	C006	Female	Grocery	Oil	5	250	
6	1007	C007	Male	Clothing	Jeans	2	1800	
7	1008	C008	Female	Electronics	Earbuds	5	1500	
8	1009	C009	Male	Furniture	Table	1	8000	
9	1010	C010	Female	Grocery	Sugar	6	80	
10	1011	C011	Male	Clothing	Jacket	2	3500	
11	1012	C012	Female	Electronics	Smart Watch	2	7000	
12	1013	C013	Male	Grocery	Wheat	4	1200	
13	1014	C014	Female	Furniture	Sofa	1	25000	
14	1015	C015	Male	Electronics	Camera	1	42000	
15	1016	C016	Female	Clothing	Dress	3	2200	
16	1017	C017	Male	Grocery	Tea	7	350	
17	1018	C018	Female	Electronics	Tablet	2	22000	
18	1019	C019	Male	Furniture	Cupboard	1	18000	
19	1020	C020	Female	Clothing	Skirt	4	1500	
20	1021	C021	Male	Electronics	Printer	2	8500	
21	1022	C022	Female	Grocery	Milk	10	60	
22	1023	C023	Male	Furniture	Bed	1	30000	
23	1024	C024	Female	Clothing	Saree	2	4000	
24	1025	C025	Male	Electronics	Monitor	3	12000	
25	1026	C026	Female	Grocery	Biscuits	8	100	
26	1027	C027	Male	Clothing	Shoes	2	3000	
27	1028	C028	Female	Furniture	Dining Table	1	35000	
28	1029	C029	Male	Electronics	Keyboard	4	1200	
29	1030	C030	Female	Grocery	Coffee	5	450	

	Sales	Profit
0	55000	7000
1	2400	500
2	3000	400
3	10000	1800
4	36000	5000
5	1250	200
6	3600	700
7	7500	1200
8	8000	1500
9	480	100
10	7000	1200
11	14000	2500
12	4800	600
13	25000	5000
14	42000	6500
15	6600	1000
16	2450	350
17	44000	6000
18	18000	3200
19	6000	900
20	17000	2500
21	600	120
22	30000	5500

```

23    8000    1500
24   36000    4500
25     800     160
26    6000     900
27   35000    6000
28    4800     800
29    2250     300

```

```
In [12]: df.shape
```

```
Out[12]: (30, 9)
```

```
In [3]: df.head(8)
```

```
Out[3]:
```

	Order_ID	Customer_ID	Gender	Category	Product	Quantity	Price	Sales	Profit
0	1001	C001	Male	Electronics	Laptop	1	55000	55000	7000
1	1002	C002	Female	Clothing	T-Shirt	3	800	2400	500
2	1003	C003	Male	Grocery	Rice Bag	2	1500	3000	400
3	1004	C004	Female	Furniture	Chair	4	2500	10000	1800
4	1005	C005	Male	Electronics	Mobile	2	18000	36000	5000
5	1006	C006	Female	Grocery	Oil	5	250	1250	200
6	1007	C007	Male	Clothing	Jeans	2	1800	3600	700
7	1008	C008	Female	Electronics	Earbuds	5	1500	7500	1200

```
In [4]: df.tail()
```

```
Out[4]:
```

	Order_ID	Customer_ID	Gender	Category	Product	Quantity	Price	Sales	Profit
25	1026	C026	Female	Grocery	Biscuits	8	100	800	160
26	1027	C027	Male	Clothing	Shoes	2	3000	6000	900
27	1028	C028	Female	Furniture	Dining Table	1	35000	35000	6000
28	1029	C029	Male	Electronics	Keyboard	4	1200	4800	800
29	1030	C030	Female	Grocery	Coffee	5	450	2250	300

```
In [5]: df.columns.tolist()
```

```
Out[5]: ['Order_ID',  
        'Customer_ID',  
        'Gender',  
        'Category',  
        'Product',  
        'Quantity',  
        'Price',  
        'Sales',  
        'Profit']
```

```
In [8]: max_sales = df.groupby('Category')['Sales'].mean()  
print(max_sales)
```

```
Category  
Clothing      5657.142857  
Electronics   28477.777778  
Furniture     21000.000000  
Grocery       1953.750000  
Name: Sales, dtype: float64
```

```
In [11]: min_sales = df.groupby('Category')['Sales'].mean()  
print(min_sales)
```

```
Category  
Clothing      5657.142857  
Electronics   28477.777778  
Furniture     21000.000000  
Grocery       1953.750000  
Name: Sales, dtype: float64
```

```
In [13]: product_count = df['Category'].value_counts()  
print(product_count)
```

```
Category  
Electronics    9  
Grocery        8  
Clothing       7  
Furniture      6  
Name: count, dtype: int64
```

```
In [14]: gender_count = df['Gender'].value_counts()  
print(gender_count)
```

```
Gender  
Male    15  
Female  15  
Name: count, dtype: int64
```

```
In [16]: profit_category = df.groupby('Category')['Profit'].mean()  
print(profit_category)
```

```
Category  
Clothing      957.142857  
Electronics   4000.000000  
Furniture     3833.333333  
Grocery       278.750000  
Name: Profit, dtype: float64
```

```
In [17]: sales_category = df.groupby('Category')['Sales'].mean()
print(sales_category)
```

```
Category
Clothing      5657.142857
Electronics   28477.777778
Furniture     21000.000000
Grocery       1953.750000
Name: Sales, dtype: float64
```

```
In [18]: avg_sales = df.groupby('Gender')['Sales'].mean()
print(avg_sales)
```

```
Gender
Female      10925.333333
Male        18243.333333
Name: Sales, dtype: float64
```

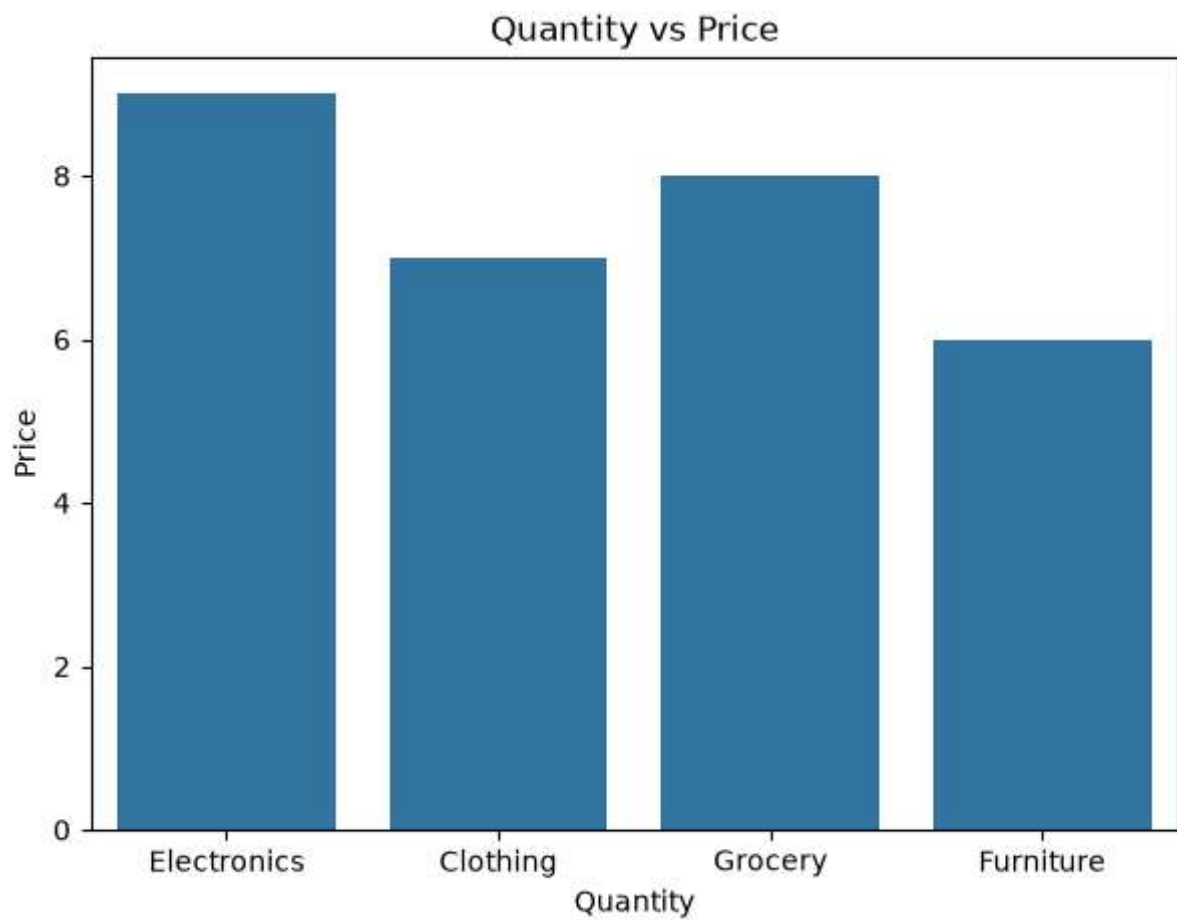
```
In [20]: highest_category = df.groupby('Category')['Sales'].mean()
print("Highest Selling Product Category:", highest_category)
```

```
Highest Selling Product Category: Category
Clothing      5657.142857
Electronics   28477.777778
Furniture     21000.000000
Grocery       1953.750000
Name: Sales, dtype: float64
```

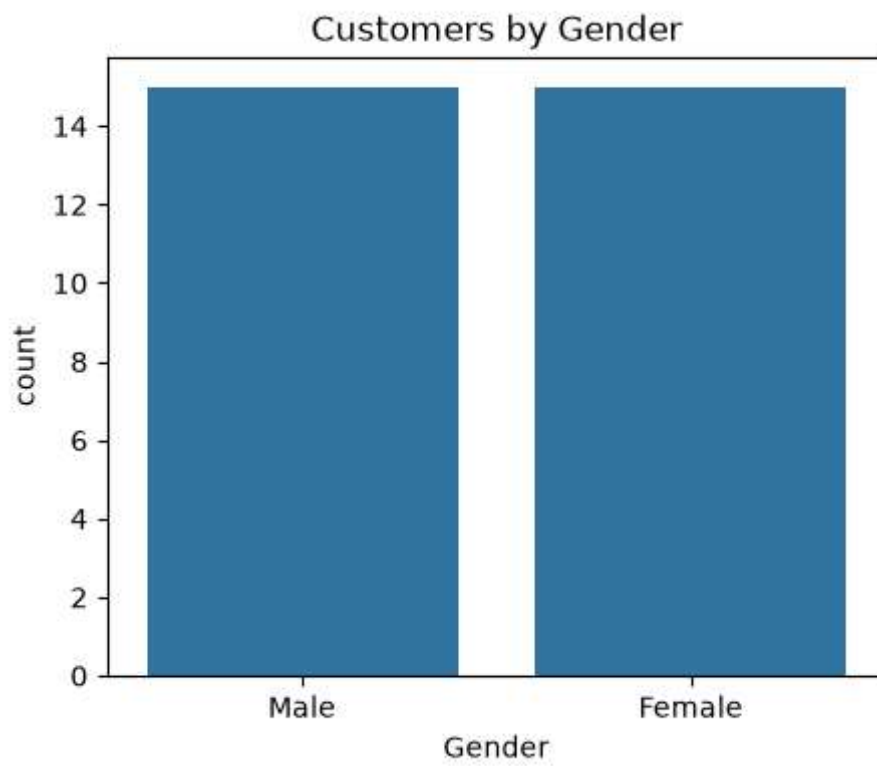
```
In [22]: correlation=df['Quantity'].corr(df['Price'])
print("Correlation between Quantity and Price:", correlation)
```

```
Correlation between Quantity and Price: -0.574399955304757
```

```
In [31]: plt.figure(figsize=(7,5))
sns.countplot(x='Category', data=df)
plt.title("Quantity vs Price")
plt.xlabel("Quantity")
plt.ylabel("Price")
plt.show()
```



```
In [32]: plt.figure(figsize=(5,4))
sns.countplot(x='Gender',data=df)
plt.title("Customers by Gender")
plt.show()
```



In []: